

CEVE 421/521 Schedule

i Note

Check the page for each reading to see discussion questions. You will often be encouraged to focus only on part of relevant papers.

Final Project: Deliverables are due throughout the semester (see schedule below).

CEVE 521 students: You will lead one Paper Discussion session. See Graduate Seminar Requirements.

Week	Date	What
1	Mon., Jan. 12	Lecture: Intro to Climate Risk Management slides
	Wed., Jan. 14	Topical Discussion: T. Frank [1] and S. Crawford [2]
	Fri., Jan. 16	Lab 1: Julia Setup & Hello World
2	Mon., Jan. 19	<i>Martin Luther King, Jr. Day (No Class)</i>
	Wed., Jan. 21	Lecture: Hazard, Exposure, Vulnerability slides
	Fri., Jan. 23	Lab 2: Intro to iCOW
3	Mon., Jan. 26	Lecture: Climate Variability, Extremes, and Change slides (<i>in prep</i>)
	Wed., Jan. 28	Paper Discussion: Climate Extremes (IPCC) (<i>in prep</i>)
	Fri., Jan. 30	Lab 3: Sea-Level Rise Scenarios with BRICK (<i>in prep</i>) / Project: Topic Proposal Due
4	Mon., Feb. 2	Lecture: Uncertainty & Monte Carlo slides (<i>in prep</i>)
	Wed., Feb. 4	Paper Discussion: Monte Carlo & Fat Tails (<i>in prep</i>)
	Fri., Feb. 6	Lab 4: Monte Carlo Hazard Modeling (<i>in prep</i>)
5	Mon., Feb. 9	Lecture: Risk Metrics (EAD, VaR, CVaR) slides (<i>in prep</i>)
	Wed., Feb. 11	Lab 5: Probabilistic Risk Analysis (<i>in prep</i>)
	Fri., Feb. 13	<i>No Class</i>

Week	Date	What
6	Mon., Feb. 16	Exam 1 (Covers Module 1)
	Wed., Feb. 18	Lecture: Valuation & Benefit-Cost Analysis slides (<i>in prep</i>)
	Fri., Feb. 20	Paper Discussion: Discounting & BCA (<i>in prep</i>)
7	Mon., Feb. 23	Lecture: Optimal Static Decisions slides (<i>in prep</i>)
	Wed., Feb. 25	Paper Discussion: Optimal Decisions (<i>in prep</i>)
	Fri., Feb. 27	Lab 6: Simulation-Optimization (<i>in prep</i>) / Project: Memo 1 Due
8	Mon., Mar. 2	Lecture: Sensitivity & Value of Information slides (<i>in prep</i>)
	Wed., Mar. 4	Paper Discussion: Value of Information (<i>in prep</i>)
	Fri., Mar. 6	Lab 7: Value of Partial Information (<i>in prep</i>)
9	Mon., Mar. 9	Lecture: Trade-offs & Values slides (<i>in prep</i>)
	Wed., Mar. 11	Paper Discussion: Trade-offs and Values (<i>in prep</i>)
	Fri., Mar. 13	Exam 2 (Covers Module 2)
	Mar. 16-20	<i>Spring Break</i>
10	Mon., Mar. 23	Lecture: Robustness & Deep Uncertainty slides (<i>in prep</i>)
	Wed., Mar. 25	Paper Discussion: Robustness & Deep Uncertainty (<i>in prep</i>)
	Fri., Mar. 27	Lab 8: Robustness Metrics (<i>in prep</i>) / Project: Memo 2 Due
11	Mon., Mar. 30	Lecture: Exploratory Modeling & Scenario Discovery slides (<i>in prep</i>)
	Wed., Apr. 1	Paper Discussion: Exploratory Modeling (<i>in prep</i>)
	Fri., Apr. 3	Lab 9: Scenario Discovery (<i>in prep</i>)
12	Mon., Apr. 6	Lecture: Sequential Decision Making slides (<i>in prep</i>)
	Wed., Apr. 8	Paper Discussion: Sequential Decision Making (<i>in prep</i>)

Week	Date	What
13	Fri., Apr. 10	Lab 10: Real Options & Dynamic Policy Search (<i>in prep</i>) / Project: Memo 3 Due
	Mon., Apr. 13	Lecture: Multi-Objective Optimization slides (<i>in prep</i>)
	Wed., Apr. 15	Paper Discussion: Multi-objective Optimization (<i>in prep</i>)
14	Fri., Apr. 17	Lab 11: Pareto Fronts (<i>in prep</i>) / Project: Slides Due
	Mon., Apr. 20	Final Project Presentations (Teams 1 & 2)
	Wed., Apr. 22	Final Project Presentations (Teams 3 & 4)
	Fri., Apr. 24	Final Project Presentations (Teams 5 & 6)

Bibliography

- [1] T. Frank, "Bold New Jersey Shore Flood Rules Could Be Blueprint for Entire U.S. Coast," *Scientific American*, Aug. 2022, Accessed: Jan. 11, 2023. [Online]. Available: <https://www.scientificamerican.com/article/bold-new-jersey-shore-flood-rules-could-be-blueprint-for-entire-u-s-coast/>
- [2] S. Crawford, "Floods and Storms Are Ravaging the Jersey Shore. Why Do We Keep Building It Back?." Accessed: Dec. 05, 2025. [Online]. Available: <https://www.rollingstone.com/culture/culture-features/floods-storms-jersey-shore-beach-rebuilding-1235477927/>